

THEORY AND METHODS
APPLICATION OF A BIVARIATE LATENT
TRAIT MODEL

J.S. Maritz

Biostatistics Unit of the South African Medical Research Council

and

S. Dyer

Department of Obstetrics and Gynaecology, University of Cape Town

Key words: Correlated data; latent traits.

Summary: A latent trait model for correlated binary data is proposed and applied to data obtained in a study of parenthood motives.

1. Introduction

In a study of motives for wanting a child, 50 infertile couples were asked to complete the parenthood-motivation list, an instrument developed in the Netherlands for the assessment of parenthood motives (van Balen & Trimbos-Kemper, 1995). Each member of the couple was asked to respond separately to the same set of questions. The section of the questionnaire relevant to this paper comprises 18 questions, referred to as questions A, B,...,R. According to the instrument the questions form 6 groups of 3 questions each, each group for a specific factor (i.e. parenthood motive). The responses to the questions are captured on a 3-point categorical scale. While modelling 3-point categorical responses is possible dealing with binary responses is much easier. In most cases one of the categories was dominant, and in consultation with the researcher the 3-point scale was reduced to a binary scale by amalgamating

two of the categories. For the illustration in this paper the group of questions C,H,M is used. The results obtained for the other groups are similar to those reported here for C,H,M.

One of the objectives in the analysis of the data was to see whether the factor model proposed by van Balen & Trimbos-Kemper (1995) is applicable in the South African context. A straightforward approach is simply to consider each triplet of questions defining a group separately and to check if a one-factor model adequately explains the relationships between the three variables defined by the questions.

2. A binary response latent trait model

In preparation for the model proposed for our data we give a brief description of standard latent trait modelling of binary response data; see for example Bartholomew and Knott (1999). Consider n subjects each of whom gives a Yes/No response to a number, J , of questions. The underlying idea in latent trait modelling is that the probability of subject i answering “Yes” to question j is determined by that subject’s values, $\theta'_i = (\theta_{1i}, \theta_{2i}, \dots, \theta_{Ti})$, of $T < J$ variables that are not directly observed, or observable. They are the *latent traits*.

Let $Y_{ij} = 1$ if the response by subject i to question j is “Yes”, else $Y_{ij} = 0$. Then

$$P(Y_{ij} = 1 \mid \theta_i) = g_j(\theta_i).$$

Important assumptions:

(1)

$$P(Y_{ij} = 1, Y_{ik} = 1 \mid \theta_i) = g_j(\theta_i)g_k(\theta_i);$$

it is referred to as conditional independence of questions j and k within a subject as specified by a T-vector of traits.

- (2) As usual, also, the responses of the subjects are assumed to be independent.
- (3) In the population of subjects the latent traits are all independent and each is distributed $N(0, 1)$.

What remains before we can apply these ideas in the analysis of data is specification of the functions $g_j(\boldsymbol{\theta})$.

The logistic model

$$\ln \left(\frac{P(Y_{ij} = 1 | \boldsymbol{\theta}_i)}{1 - P(Y_{ij} = 1 | \boldsymbol{\theta}_i)} \right) = \alpha_j + \beta_{1j}\theta_{1i} + \beta_{2j}\theta_{2i}, \dots, \beta_{Tj}\theta_{Ti} = \alpha_j + \boldsymbol{\beta}'_j \boldsymbol{\theta}_i, \quad (1)$$

where $\boldsymbol{\beta}'_j = (\beta_{1j}, \beta_{2j}, \dots, \beta_{Tj})$, guarantees that the modelled probabilities lie between 0 and 1. We can write this as

$$\begin{aligned} P(Y_{ij} = 1 | \boldsymbol{\theta}_i) &= \exp(\alpha_j + \boldsymbol{\beta}'_j \boldsymbol{\theta}_i) / [1 + \exp(\alpha_j + \boldsymbol{\beta}'_j \boldsymbol{\theta}_i)] \\ &= F(\alpha_j + \boldsymbol{\beta}'_j \boldsymbol{\theta}_i). \end{aligned} \quad (2)$$

The Rasch model is obtained by putting $\boldsymbol{\beta}'_j = (\beta_1, \beta_2, \dots, \beta_T) = \boldsymbol{\beta}'$ in (1). A more flexible version of (1) is obtained by adding integral powers of the θ_j to the r.h.s. of the equation; for example, one might have

$$\ln \left(\frac{P(Y_{ij} = 1 | \boldsymbol{\theta}_i)}{1 - P(Y_{ij} = 1 | \boldsymbol{\theta}_i)} \right) = \alpha_j + \beta_{11j}\theta_{1i} + \beta_{12j}\theta_{1i}^2 + \beta_{2j}\theta_{2i}. \quad (3)$$

Thus the example in (3) is a special case of (2) where $\boldsymbol{\theta}_i$ may contain powers of individual θ 's.

3. Observations in pairs

A typical case is where the respondents to questions are the members of a married couple. In theory the females may be thought of as having values of latent traits $\boldsymbol{\theta}_F$, the males $\boldsymbol{\theta}_M$. Considering J questions as before, we now have in an obvious extension of the earlier notation

$$P(Y_{Fij} = 1 | \boldsymbol{\theta}_{Fi}) = F(\alpha_{Fj} + \boldsymbol{\beta}'_{Fj} \boldsymbol{\theta}_{Fi}) = P_{Fij}.$$

for the female responses and similarly for male responses.

Now, particular possible dependencies have to be considered.

- (1) Conditional independence of responses to different questions within the same female or male subject, given θ_F and θ_M : We assume that

$$P(Y_{Fij} = 1, Y_{Fik} = 1 | \theta_{Fi}) = P(Y_{Fij} = 1 | \theta_{Fi}) P(Y_{Fik} = 1 | \theta_{Fi})$$

and likewise for males. This assumption is the same as the conditional independence assumed before.

- (2) Conditional independence of responses to different questions by members of the same pair given θ_F and θ_M : We assume

$$P(Y_{Fij} = 1, Y_{Mik} = 1 | \theta_{Fi}, \theta_{Mi}) = P(Y_{Fij} = 1 | \theta_{Fi}) P(Y_{Mik} = 1 | \theta_{Mi}).$$

This is similar to assumption (1) above, but somewhat weaker.

- (3) The responses to the same question by members of the same pair may not be independent. Suppose that the correlation of the Bernoulli variables Y_{Fij} and Y_{Mij} is ρ_{ij} . For fixed $P_{Fij} = P(Y_{Fij} = 1)$ and $P_{Mij} = P(Y_{Mij} = 1)$ the maximum value of ρ_{ij} is not necessarily +1, but occurs at the maximum value of $P(Y_{Fij} = 1, Y_{Mij} = 1 | \theta_{Fi}, \theta_{Mi})$ which is $Q_{ij}^{\max} = \min(P_{Fij}, P_{Mij})$. Consequently we model positive dependence by a parameter ψ_{ij} , where $0 \leq \psi_{ij} < 1$, and let

$$P(Y_{Fij} = 1, Y_{Mij} = 1 | \theta_{Fi}, \theta_{Mi}) = P_{Fij} P_{Mij} (1 - \psi_{ij}) + \psi_{ij} Q_{ij}^{\max}. \quad (4)$$

The case of negative dependence is dealt with similarly. The greatest negative correlation occurs at the minimum value of $P(Y_{Fij} = 1, Y_{Mij} = 1 | \theta_{Fi}, \theta_{Mi})$, i.e. $Q_{ij}^{\min}$. Now

$$Q_{ij}^{\min} = \max[Q_{ij}^{\min 1}, Q_{ij}^{\min 2}],$$

where

$$\begin{aligned} Q_{ij}^{\min 1} &= \min(P_{Fij}, P_{Mij}) - \min(1 - P_{Fij}, P_{Mij}) \\ Q_{ij}^{\min 2} &= \min(P_{Fij}, P_{Mij}) - \min(P_{Fij}, 1 - P_{Mij}) \end{aligned}$$

and, for $-1 < \psi_{ij} \leq 0$, we let

$$P(Y_{Fij} = 1, Y_{Mij} = 1 | \theta_{Fi}, \theta_{Mi}) = P_{Fij} P_{Mij} (1 + \psi_{ij}) - \psi_{ij} Q_{ij}^{\min}. \quad (5)$$

To make progress with this modelling, it is clear that some simplifying

assumptions have to be made to deal with the large number of ψ_{ij} . Two obvious possibilities are

- (a) $\psi_{ij} = \psi$
 - (b) $\psi_{ij} = 0$ for all i and j .
- (4) The marginal distributions of all latent traits are taken to be $N(0, 1)$. Allowance for correlation between particular latent traits may have to be made; just how will depend on the particular data being modelled.

4. A model for three questions

As suggested in the Introduction we consider triplets of questions and fit a model with the view to examining the propositions that

- (a) a single latent trait can explain the data, and,
- (b) there is correlation between the members of a pair.

Regarding (a) above, it was decided to examine a model in which the same latent trait controls the responses of the female and male partners in a couple. This is in effect assuming that there is a single female latent trait, θ_F , and a single male latent trait, θ_M but that they are perfectly correlated. After preliminary examination of several simple models the following was decided upon: For Question j , where $j = 1, 2, 3$,

$$\begin{aligned} P(Y_{Fij} = 1 \mid \theta_{Fi}) &= \alpha_{Fj} + \beta_{1Fj}\theta_i + \beta_{2Fj}\theta_i^2 \\ P(Y_{Mij} = 1 \mid \theta_{Mi}) &= \alpha_{Mj} + \beta_{1Mj}\theta_i + \beta_{2Mj}\theta_i^2 \end{aligned} \quad (6)$$

and all $\psi_{ij} = \psi$.

Estimation of the 19 parameters is by maximum likelihood, or to be more precise, by maximizing the expected, or integrated likelihood, the integration being performed over the θ_i . Suppose, for example, that the observed values for couple i on Question j are $Y_{Fij} = 1$ and that ψ is negative. Then the likelihood of this observation is given by the r.h.s. of equation (5) with

$\psi_{ij} = \psi$, i.e.

$$Q_i = P_{Fij}P_{Mij}(1 + \psi) - \psi Q_{ij}^{\min}.$$

If observed $Y_{Fij} = 0$ and $Y_{Mij} = 1$ then the likelihood is $P_{Mij} - Q_i$, etc.

In this way a likelihood is obtained for each question, and their product is the likelihood for couple i . Next we integrate this likelihood w.r.t. θ_i , recalling that it is taken to be distributed $N(0, 1)$. This calculation is performed for every couple and we then obtain the sum, $\ln L$, of the logarithms of these integrated likelihoods. The last step is to maximize $\ln L$ w.r.t. the parameters $\alpha_{F1}, \dots, \beta_{M3}, \psi$.

5. Application to the parenthood motive data

Motive: Well-being Questions: C, H, M

Table 1 shows the data for the 50 couples in the study.

Fitting model (6) with ψ not necessarily zero, the maximum $\ln L = -171.35$ and it is at $\psi = -0.273$. Setting $\psi = 0$ gives maximum $\ln L = -171.90$.

The goodness of fit of the model can be examined by calculating the expected frequency of every one of the 64 possible configurations of which examples are shown in Table 1, using the estimated parameters. A goodness-of-fit statistic is $Q = \Sigma(O - E)^2/E$ where O and E are the observed and expected frequencies. With the total sample size $n = 50$ and there being 64 configurations, it is obvious that the chi-squared approximation of the distribution of Q cannot be expected to be good. However, it may be reasonable to assume that the expected value of Q is $64 - 1 - k$, where k is the number of parameters fitted. The observed value of Q with maximum likelihood estimates of 18 parameters and $\psi = 0$ is 52.05. The observed

value of Q with the maximum likelihood estimates of the model with non-zero ψ , i.e. 19 fitted parameters, is 46.97.

Table 1.

Couple	Cf	Cm	Hf	Hm	Mf	Mm	Couple	Cf	Cm	Hf	Hm	Mf	Mm
1	1	1	1	1	1	1	26	1	0	1	0	0	0
2	0	1	1	1	1	1	27	0	1	0	1	0	0
3	0	1	1	1	0	1	28	1	0	1	1	0	0
4	1	0	0	1	1	1	29	1	1	1	1	1	1
5	1	1	1	1	1	1	30	1	1	1	1	1	1
6	1	1	1	1	0	1	31	1	1	1	1	1	1
7	1	1	0	1	1	1	32	1	1	1	1	1	1
8	0	0	0	1	0	0	33	1	1	1	0	1	1
9	1	1	1	1	1	1	34	1	0	1	0	1	0
10	1	1	0	1	1	1	35	0	1	1	0	1	0
11	1	0	1	0	0	1	36	0	0	1	1	1	0
12	1	1	1	0	1	1	37	0	0	0	0	0	1
13	1	1	1	1	1	1	38	1	1	1	1	1	1
14	1	0	1	1	1	1	39	0	0	0	0	0	0
15	1	0	1	0	1	1	40	1	0	0	0	1	0
16	0	0	0	0	1	1	41	1	1	0	0	0	1
17	1	1	1	1	1	1	42	1	0	1	1	1	0
18	0	0	1	0	0	0	43	0	0	0	0	0	0
19	1	1	1	1	1	1	44	0	0	0	0	0	1
20	1	1	1	0	0	0	45	1	0	1	0	1	0
21	0	0	0	1	1	0	46	1	0	1	0	1	1
22	1	1	1	1	1	1	47	1	0	1	0	1	1
23	0	0	1	1	1	0	48	1	0	0	0	0	1
24	0	1	1	1	0	1	49	0	0	0	0	0	0
25	1	1	1	1	1	0	50	1	0	0	0	0	0

For Figure 1 the expected frequencies, with $\psi = -0.273$, were sorted from smallest to largest, the cumulative frequencies calculated, along with the corresponding observed frequencies. Observed cumulative frequencies are plotted against expected cumulative frequencies. For a satisfactory fit the points should cluster around the straight line shown in the graph. The curves

in Figure 1 are approximate 5 and 95 percentiles for the observed cumulative frequencies.

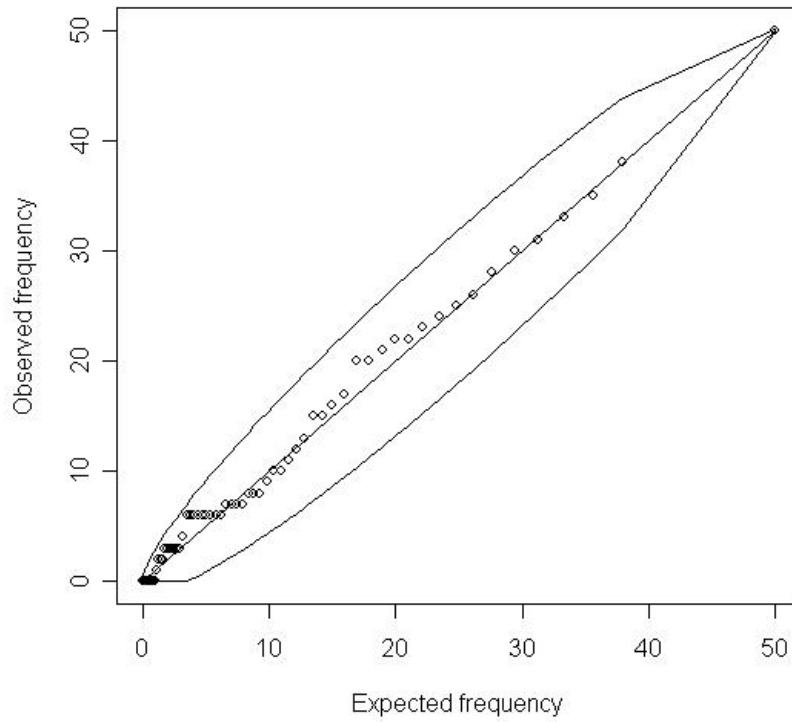


Figure 1. Data = CHM

Conclusions

- (a) The assessment of goodness of fit of model (6) indicates that the hypothesis of a single latent trait accounting for the pattern of responses of the couples to questions C,H,M is acceptable.
- (b) The difference between the maximum $\ln L$ obtained with $\psi = 0$ and $\psi \neq 0$ is small, hence the hypothesis that the female and male responses are independent can be accepted.

6. Computational note

All calculations were performed by programs written in R, R Development Core Team (2005). Maximizing of likelihood was performed by a Newton-Raphson search in which the partial derivatives were calculated approximately numerically, and by random search.

Fitting model (6) with $\psi = 0$ using the *ltm* package in R gives the maximum $\ln L = -171.912$. Using the parameter estimates given by the package in our calculation of $\ln L$ gives the result -171.903 . Allowing for differences that can arise especially in the numerical integration of the likelihoods, this small discrepancy is taken to confirm the correctness of our calculations.

References

- BARTHOLOMEW, D.J. & KNOTT, M. (1999). *Latent variable models and factor analysis* (2nd ed.), London: Arnold. (Kendall's Library of Statistics 7).
- VAN BALEN, F. & Trimbos-Kemper, T.C.M. (1995). Involuntarily childless couples: their desire to have children and their motives, *J.Psychosom. Obstet. Gynecol*, **16**, 137–144.
- R. DEVELOPMENT CORE TEAM (2006). R.: *A language and environment for statistical computing*, R. Foundation for Statistical Computing, Vienna, Austria. ISBN 3–900051–07–0, URL <http://www.R-project.org>.